

DSTAG: A Semantic Tag-Enhanced Dual-Graph Convolutional Network for Temporal Knowledge Graph Completion

Yuchao Zhang
Zhejiang University of Technology
College of Computer Science and
Technology
Hangzhou, China
Zhejiang Key Laboratory of Visual
Information Intelligent Processing
Hangzhou, China
yuchao.z@outlook.com

Xiangjie Kong*
Zhejiang University of Technology
College of Computer Science and
Technology
Hangzhou, China
Zhejiang Key Laboratory of Visual
Information Intelligent Processing
Hangzhou, China
xjkong@ieee.org

Kailun Ye
Zhejiang University of Technology
College of Computer Science and
Technology
Hangzhou, China
Zhejiang Key Laboratory of Visual
Information Intelligent Processing
Hangzhou, China
yekailun@zjut.edu.cn

Shangfei Zheng
Zhejiang Sci-Tech University
School of Computer Science and
Technology
Hangzhou, China
zsf@zstu.edu.cn

Guojian Shen
Zhejiang University of Technology
College of Computer Science and
Technology
Hangzhou, China
Zhejiang Key Laboratory of Visual
Information Intelligent Processing
Hangzhou, China
gjshen1975@zjut.edu.cn

Abstract

Temporal Knowledge Graph Completion (TKGC) aims to predict missing entities or relations based on historical facts, thereby facilitating the understanding of dynamic system evolution and supporting downstream reasoning tasks. However, existing methods predominantly focus on modeling sequential and structural dependencies, often overlooking the rich semantic information embedded in entities and relations, as well as the higher-order interactions among them, which limits their ability to handle complex, evolving scenarios effectively. To address these limitations, we propose **DSTAG**, a novel TKGC approach based on a semantic tag-enhanced dual-graph convolutional network. Our method leverages large language models to generate contextualized semantic multi-tags for both entities and relations (e.g., "political event," "economic activity"), thereby enriching their semantic representations. Furthermore, we introduce a semantic tag representation mechanism that captures higher-order dependencies during the aggregation and propagation of semantic tag information across graphs. DSTAG adopts a dual-graph convolutional network architecture, where the relation graph convolution extracts semantic features between temporal relationships and injects this information into the entity graph convolution, enabling joint modeling of entities and relations. We evaluate DSTAG on three widely used TKG benchmarks:

ICEWS14, ICEWS18, and ICEWS05-15. Experimental results show that DSTAG achieves substantial MRR improvements over state-of-the-art baselines by **8.64%**, **9.81%**, and **4.56%**, respectively.

CCS Concepts

• **Computing methodologies** → **Semantic networks; Temporal reasoning.**

Keywords

Temporal Knowledge Graph Completion, Dual-Graph Convolutional Network, Semantic Tagging, Large Language Models

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1 Introduction

Knowledge Graphs (KGs) are core tools for representing structured world knowledge, typically organized in the form of triplets (head entity, relation, tail entity). However, the high dynamics of the real world require knowledge representations that can capture the temporal evolution of facts. To address this, Temporal Knowledge Graphs (TKGs) have emerged. TKGs extend traditional triplets into quadruples (head entity, relation, tail entity, time), associating each fact with its timestamp or time interval, thereby precisely modeling the changing nature of facts over time [1–3]. This representation effectively characterizes facts at specific time points or periods, providing strong support for dynamic knowledge modeling.

*Corresponding authors.



TKGs have demonstrated significant application value in several fields, such as event prediction [4], time-sensitive recommender systems [5–7], and question-answering systems [8, 9]. Specifically, in event prediction, TKGs leverage historical patterns to infer future political or economic events; in recommender systems, TKGs provide personalized recommendations by capturing the temporal dynamics of user behavior; in question answering, TKGs support time-sensitive fact queries. These applications all rely on TKGs' precise modeling of dynamic entities and relationships. However, the construction of TKGs faces the common challenge of factual incompleteness. Despite ongoing improvements in KG construction efficiency and data collection capabilities, the problem of missing facts remains significant. For example, only about 30% of entities in Wikidata have a birthdate attribute [10, 11]. Temporal KGC (TKGC) aims to infer missing entities or relationships in future events using historical data, enabling enhanced representations for applications like relationship inference and temporal event prediction.

Despite the significance of the TKGC task, one of its core challenges lies in the limitations of existing methods when handling complex evolving scenarios. Current approaches primarily focus on modeling temporal and structural dependencies, often neglecting the rich semantic information embedded in entities and relations, as well as the impact of higher-order interactions among them [12]. This neglect of semantics and higher-order interactions significantly restricts the model's ability to derive comprehensive representations from data. Moreover, the semantic information of relations is crucial in TKGs, as relations not only describe interactions between entities but also carry time-varying contextual meanings (e.g., the meaning of "cooperation" may pertain to politics, economics, or culture at different times). However, existing methods often fail to fully exploit the semantic potential of relations, making it challenging to achieve accurate predictions in complex evolving scenarios.

To overcome the outlined issues, we present DSTAG, a dual-graph convolutional network method infused with semantic tags, designed to alleviate the sparsity of entities and relations in TKGs by utilizing semantic tags and modeling complex interactions between relations and entities. First, we utilize LLMs to generate semantic multi-tags for entities and relations, associating each with multiple semantic tags to enrich their semantic representations. Subsequently, we introduce a semantic tag representation method that captures higher-order dependencies among entities and among relations during the aggregation and representation of semantic tag information. Moreover, DSTAG employs a dual-graph convolutional network as the backbone for entity prediction in TKGC tasks, using relation graph convolution to extract semantic features from temporal relations and injecting these as context into entity graph convolution for joint entity-relation modeling. Experiments on the widely used TKGC benchmark datasets demonstrate that DSTAG significantly outperforms existing methods in prediction accuracy and robustness. Our primary contributions are outlined below:

- We propose DSTAG, a dual-graph convolutional network infused with semantic tags for TKGC, leveraging large language models to create semantic multi-tags for entities and relations, thereby enhancing their representational power.
- We propose a novel semantic tag representation mechanism that incorporates tag information as auxiliary semantic signals, enabling the model to capture higher-order correlations between entities and between relations and to enhance its reasoning ability.
- Extensive experiments on ICEWS14, ICEWS18, and ICEWS05-15 demonstrate that DSTAG significantly outperforms existing methods in prediction accuracy, as measured by MRR and Hits@k metrics, validating the effectiveness of our approach.

2 Related Work

TKGC methods are primarily divided into two categories [1]: one extends static knowledge graph embedding methods, and the other leverages neural network-based dynamic modeling approaches. In recent years, specialized research has also emerged focusing on adaptive feature enhancement for entities and relations in TKGs.

2.1 Extensions of Static KGC Methods

Extensions of static KGC methods typically incorporate temporal information into the embeddings of entities and relations using shallow encoders. Common strategies include distance metrics based on head and tail entity embeddings or tensor decomposition to evaluate the plausibility of facts, with optimization of the shallow encoder to endow embeddings with temporal characteristics. Representative methods include TTransE [13], ChronoR [14], T-DistMult [15], TNTComplex [16], and TeLM [17]. These approaches attempt to model the dynamic evolution of entities and relations through specialized temporal embedding techniques. However, such methods heavily rely on the optimization of randomly initialized embeddings and are constrained by limited model parameters, resulting in restricted generalization capabilities in complex dynamic scenarios.

2.2 Neural Network-Based TKGC Methods

Neural network-based TKGC methods typically combine variants of recurrent neural networks (RNNs) with graph neural networks (GNNs) [12, 18–23]. RNNs are used to model the dynamic evolution in the temporal dimension, while GNNs capture the spatial topological structure among entities, thereby generating entity representations that integrate both temporal and structural features. These approaches adeptly model the dynamic trends of entities and relations across time, as well as their sequential dependencies, making them particularly well-suited for dynamic knowledge graph scenarios. For example, RE-NET [24] employs graph convolutional networks (GCNs) to model interactions between entities, integrates RNNs to capture temporal dependencies, and predicts the probability distribution of facts through a free regression approach. TIRGN [25] further introduces multi-relational GCNs to perform differentiated modeling of entity interactions under various relation types, while using gated recurrent units (GRUs) to independently capture the dynamic progression of entities and relations, thereby enhancing the model's capacity to represent complex dynamic relations. Despite significant advancements in leveraging neighborhood information, these methods often overlook the rich semantic information embedded in entities and relations, as well as their higher-order interactions, leading to limited performance in handling complex evolving scenarios.

2.3 Feature Enhancement-Based TKGC Methods

Feature enhancement-based TKGC methods can be broadly divided into two categories. The first category leverages logical rules, by learning these rules or exploring paths related to the query, to supplement additional information for the current query, thereby effectively improving the model’s accuracy when learning entity or relation representations [26–28]. The second category focuses on enriching the representations of entities and relations through data augmentation, aiming to enhance the model’s completion performance in complex dynamic scenarios. This paper concentrates on the latter approach.

In the realm of data augmentation, Zhang et al.[20] proposed inferring latent relations by computing similarities between unconnected entities to enhance prediction accuracy. However, this approach overlooks the semantic properties of relations, which in TKGs not only represent connections between entities but also encapsulate complex semantics that evolve over time. Ding et al.[29] proposed a concept-aware TKGC model that enhances entity features by crawling relevant concept descriptions from the official dataset website. However, this approach is limited by whether the specific dataset provides such corresponding information, and it overlooks the rich interactive information among relations. Mirta-heri et al.[30] introduced a latent relation mining method based on shared links, positing that entities connected through common neighbor entities may have potential relations. While effective in specific graph structures, this method’s reliability diminishes in TKGs with complex or diverse topologies, often introducing noisy relations. Additionally, Zhang et al.[12] developed an adaptive neighborhood enhancement layer that systematically explores and supplements latent structural and semantic information based on the sparsity of entity relations. However, this approach requires examining each entity’s neighborhood individually for adaptive augmentation, which may incur significant computational overhead in large-scale TKGs.

In contrast to existing methods, we propose DSTAG, which incorporates semantic tags to enrich the semantic representations of entities and relations. By capturing higher-order semantic correlations between entities and between relations, DSTAG improves the model’s capacity to address dynamic TKG scenarios, achieving superior completion performance. Evaluation outcomes show that DSTAG surpasses current approaches in terms of prediction accuracy and stability.

3 Preliminary and Definition

In this section, we systematically elaborate on the fundamental concepts of TKGC and provide a detailed description of the core symbols, structures, and key processes involved in TKG modeling and completion tasks.

Definition 1 (TKG): A TKG is a directed graph structure that describes dynamic events in the real world, represented as a quadruple $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T}, \mathcal{F})$, where $\mathcal{E}$ is the set of entities; $\mathcal{R}$ is the set of relations; $\mathcal{T}$ is the set of timestamps; $\mathcal{F} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E} \times \mathcal{T}$ is the set of facts, where each fact $(e_s, r, e_o, t) \in \mathcal{F}$ represents an interaction between entities $e_s, e_o \in \mathcal{E}$ through relation $r \in \mathcal{R}$ at time $t \in \mathcal{T}$. A TKG can be represented as a time series $\mathcal{G} = \mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_t$, where $\mathcal{G}_t$ is the subgraph at time t containing all the facts at that moment.

Definition 2 (Entity Graph): An entity graph is constructed from the set of nodes in the event set. The entity graph at time t can be formally represented as: $\mathcal{G}_e^t = (\mathcal{V}_e^t, \mathcal{E}_e^t)$, which represents the interaction structure between entities at time t , where, $\mathcal{V}_e^t \subseteq \mathcal{E}$ is the set of entity nodes; $\mathcal{E}_e^t \subseteq \mathcal{V}_e^t \times \mathcal{V}_e^t$ is the set of edges, and an edge $(e_i, e_j) \in \mathcal{E}_e^t$ indicates that entities e_i, e_j interact at time t through some relation $r \in \mathcal{R}$.

Definition 3 (Relationship Graph): The relationship graph $\mathcal{G}_r^t(t) = (\mathcal{V}_r^t, \mathcal{E}_r^t)$ is used to describe the structural associations between relationships, where, $\mathcal{V}_r^t \subseteq \mathcal{R}$ is the set of relationship nodes involved at time t ; $\mathcal{E}_r^t \subseteq \mathcal{V}_r^t \times \mathcal{V}_r^t$ is the set of edges, and an edge $(r_i, r_j) \in \mathcal{E}_r^t$ indicates that relationships r_i and r_j share at least one entity in $\mathcal{G}_r^t$.

Definition 4 (Semantic Tag): Semantic tags are used to enrich the semantic information of entities or relations in TKGs, enhancing their expressiveness. Each entity or relation can be associated with multiple semantic tags. Formally, let $\mathcal{N}^{etag} = \{N_1^{etag}, N_2^{etag}, \dots\}$ and $\mathcal{N}^{rtag} = \{N_1^{rtag}, N_2^{rtag}, \dots\}$ denote the sets of entity and relation semantic tags, respectively.

Definition 5 (TKGC Task): The TKGC task aims to complete the absent facts in a TKG based on historical facts. It is categorized into interpolation and extrapolation approaches. In the interpolation mode, missing facts within time $\mathcal{T}$ are completed based on existing facts $\mathcal{F}$. In the extrapolation mode, historical facts $\mathcal{F}_{[:t-1]}$ are used to predict facts at a future time t , formalized as $(e_s, r, ?, t)$ or $(?, r, e_o, t)$. This paper focuses on the extrapolation task, with the goal of predicting the set of facts $\mathcal{F}_t$ in $\mathcal{G}_t$ given $\mathcal{G}_{[:t-1]} = \mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_{t-1}$.

4 Methodology

As shown in Figure 1, the proposed DSTAG method enhances entity and relation representations in TKG at each temporal snapshot by enriching their embeddings and capturing higher-order interactions among entities and among relations. DSTAG consists of two main modules: the Dual-Graph Convolutional Network (DGCN) and the Semantic Tag Augmentation Network (STAN). The latter is further divided into the LLM Semantic Tags generation submodule (LLM-ST), Semantic Tag Representation submodule (STR), and the Semantic Message Passing submodule (SMP). In the DGCN module, DSTAG constructs separate entity and relation graphs, and employs a dual-graph convolutional network to capture semantic interactions among entities and among relations. In the STAN module, a LLM is used to discover semantic associations between entities and between relations, which are then represented using the STR submodule. Subsequently, the SMP submodule updates the features of entity and relation nodes through a message-passing mechanism to reflect semantic features via backpropagation.

4.1 DGCN

To capture the spatial and semantic dependencies among facts at each timestamp, inspired by [31–33], we construct entity and relation graphs from the fact sets at each timestamp. A DGCN serves as the backbone for entity prediction in our TKGC model. While modeling facts at each timestamp, semantic interactions captured in the relation graph are injected into the entity graph as supplementary information to enhance entity representations. The architecture

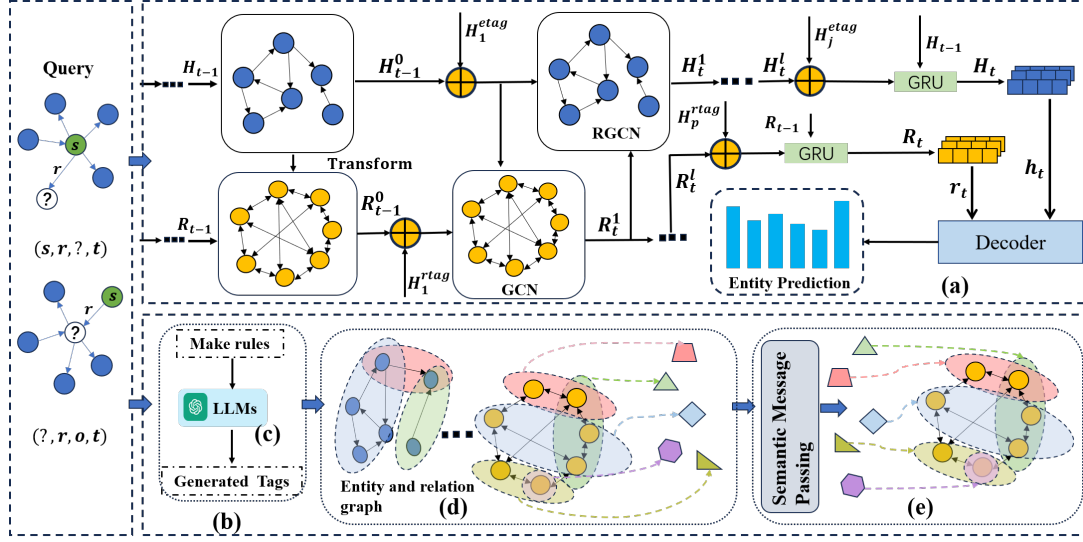


Figure 1: Overview of the proposed DSTAG model architecture. Part (a) illustrates the dual-graph convolutional network, where separate entity and relation graphs are constructed to model semantic interactions. Semantic features from the relation graph are injected into the entity graph to enhance entity representations. Parts (b)–(d) depict the semantic tag generation and enhancement module: descriptive semantics of entities and relations are extracted using a large language model, which generates semantic tags. These tags are then integrated with graph-based representations to further refine entity and relation embeddings, ultimately improving entity prediction performance.

is illustrated in Figure 2(a). Furthermore, many real-world events exhibit periodic patterns, such as academic conferences and corporate annual meetings. We follow the approach proposed in [19, 34], which integrates cyclic temporal features into the representations: Considering the time interval Δt , the periodic time function is expressed as follows:

$$\varphi(\Delta t) = \cos(\Delta t w_t + b_t) \quad (1)$$

where w_t is a learnable scaling parameter, and b_t is the bias at time t , used to capture the periodicity of temporal features. The dynamic entity embedding $\mathbf{h}_t$ is thus formulated as:

$$\mathbf{e}_t = \mathbf{W}_0 [\mathbf{e}_t^0 \parallel \varphi(\Delta t)] \quad (2)$$

where $\mathbf{e}_t^0$ denotes the initial entity embedding at time t , $\mathbf{W}_0$ is a learnable parameter matrix, and $\parallel$ represents the vector concatenation operation.

To extract contextual semantics from the relation graph $\mathcal{G}_r^t$, DGCN applies a Graph Convolutional Network (GCN) to update the node representations. Given the initial representation $\mathbf{r}_{i,t}^{(0)}$ of the relation node $r_i \in \mathcal{V}_r^t$, the GCN iteratively updates as follows:

$$\mathbf{r}_{i,t}^{(l+1)} = \sigma \left(\sum_{r_j \in \mathcal{N}_{r_i,t}} \frac{1}{|\mathcal{N}_{r_i,t}|} \mathbf{W}_1^{(l)} \mathbf{r}_{j,t}^{(l)} + \mathbf{W}_2^{(l)} \mathbf{r}_{i,t}^{(l)} \right) \quad (3)$$

where $\mathcal{N}_{r_i,t}$ is the set of neighbor nodes of r_i , $\mathbf{W}_1^{(l)}, \mathbf{W}_2^{(l)}$ are the learnable parameters of the l -th layer, and σ is the activation function ReLU. The final node representation $\mathbf{r}_{i,t}^{(l+1)}$ integrates the semantics dependencies among relations in $\mathcal{G}_r^t$, which is used for subsequent TKG prediction tasks.

To model structural and semantics dependencies in the entity graph, we adopt the method proposed in [19] for updating entity representations, which we omit here for brevity. After obtaining entity and relation representations at each timestamp, we employ a Gated Recurrent Unit (GRU) to capture their sequential dependencies. The temporal evolution of entity embeddings is formulated as:

$$\mathbf{E}_t = \text{GRU}(\mathbf{E}_{t-1}, \mathbf{E}_{t-1}^{\text{GCN}}) \quad (4)$$

where $\mathbf{E}_t$ and $\mathbf{E}_{t-1}$ denote the entity embedding matrices at times t and $t-1$, respectively, and $\mathbf{E}_{t-1}^{\text{GCN}}$ represents the entity embeddings at time $t-1$ after aggregation via entity graph.

For the evolution of relations over time, we follow the approach in [18, 19] and consider the entity embeddings $\mathbf{e}_{r,t}$ related to relation r :

$$\mathbf{r}'_t = [\text{pooling}(\mathbf{E}_{t-1}, \mathbf{e}_{r,t}); \mathbf{r}] \quad (5)$$

$$\mathbf{R}_t = \mathbf{g}_{R_t} \cdot \mathbf{R}_t + (1 - \mathbf{g}_{R_t}) \mathbf{R}_{t-1} \quad (6)$$

where $\mathbf{g}_{R_t}$ is a learnable gating vector, and $\mathbf{R}'_t$ contains all the representations of $\mathbf{r}'_t$ at time t . This mechanism effectively captures the dynamic evolution of relation representations over time.

4.2 STAN

STAN is the core innovative module of DSTAG, designed to alleviate the prevalent issues of data sparsity and semantic deficiency in TKGs by introducing external semantic priors. Unlike approaches

that rely solely on text encoders to extract implicit semantics, STAN explicitly constructs and models semantic multi-tags for entities and relations, and deeply integrates them into the temporal representation learning process through dedicated representation learning and message passing mechanisms. STAN consists of three sub-modules: LLM-ST, STR, and SMP. Its overall architecture is illustrated in Figure 1(b).

4.2.1 LLM-ST. As shown in Figure 1(c), the LLM-ST module is designed to generate tags for entities and relationships in the TKG. Specifically, the LLM-ST module employs ChatGPT-4o to produce a set of semantic tags for each entity $e_i \in \mathcal{E}$ and each relation $r_i \in \mathcal{R}$, thereby enriching facts with additional semantic information. The input instruction to ChatGPT-4o is formulated as follows:

Generate semantic tags using LLM

You are an expert in the field of TKGs. For the given action/event entities and relations, generate semantic category tags capturing different aspects, and from multiple dimensions assign one or more of the most appropriate semantic labels to each such entity/relation.
 Entities = {Leader, Organization, ...};
 Relations = {Make statement, Consult, Make an appeal or request, ...};

The GPT generates a set of semantic tags for each entity and relation (e.g., "Military" and "Government" as tags for "Military Personnel (Somalia)"). Each entity e_i and each relation r_i is associated with a set of semantic tags, denoted as $N_j^{etag} \subseteq \mathcal{N}^{etag}$ and $N_p^{rtag} \subseteq \mathcal{N}^{rtag}$, Where $\mathcal{N}^{etag}$ and $\mathcal{N}^{rtag}$ denote the global tag spaces for entities and relations, respectively. It is important to emphasize that this process does not depend on any graph structure, temporal information, or factual triples. The generated semantic tags serve as time-invariant semantic priors to compensate for the insufficient structural signals in sparse scenarios, rather than deliberately modeling the evolution of semantics over time.

4.2.2 STR. To represent semantic tags, inspired by [4, 21, 29], the STR module constructs initial tag representations based on the initial embeddings of entities and relations:

$$\mathbf{h}_j^{etag} = \frac{1}{|N_j^{etag}|} \sum_{e_i \in N_j^{etag}} \mathbf{e}_i \quad (7)$$

$$\mathbf{h}_p^{rtag} = \frac{1}{|N_p^{rtag}|} \sum_{r_i \in N_p^{rtag}} \mathbf{r}_i \quad (8)$$

To capture the contribution differences of different entities or relations to the tag representations, STR introduces an attention mechanism to refine the tag representations:

$$\mathbf{h}_j^{etag} = \sum_{e_i \in N_j^{etag}} \alpha_j^{e_i} \mathbf{e}_i, \quad \alpha_j^{e_i} = \frac{\exp(\mathbf{e}_i^\top \mathbf{h}_j^{etag})}{\sum_{e_i \in N_j^{etag}} \exp(\mathbf{e}_i^\top \mathbf{h}_j^{etag})} \quad (9)$$

$$\mathbf{h}_p^{rtag} = \sum_{r_i \in N_p^{rtag}} \alpha_p^{r_i} \mathbf{r}_i, \quad \alpha_p^{r_i} = \frac{\exp(\mathbf{r}_i^\top \mathbf{h}_p^{rtag})}{\sum_{r_i \in N_p^{rtag}} \exp(\mathbf{r}_i^\top \mathbf{h}_p^{rtag})} \quad (10)$$

where $\mathbf{h}_j^{etag}$ and $\mathbf{h}_p^{rtag}$ represent the final entity and the relation tag representation, respectively, $\alpha_j^{e_i}$ and $\alpha_p^{r_i}$ represent the relevance of the entity and relation representations to the entity tag j and relation tag p .

4.2.3 SMP. To capture the influence of semantic tags on entities and relations, the SMP module uses an attention mechanism and a Multi-Layer Perceptron (MLP) to aggregate the semantic tag representations, generating enhanced entity and relation representations. The specific implementation steps are as follows:

For an entity e_i and relation r_i , STR respectively aggregates their associated tag representations $\mathbf{h}_j^{etag}$ and $\mathbf{h}_p^{rtag}$ via an attention mechanism, yielding enhanced representations $\hat{\mathbf{e}}_i$ and $\hat{\mathbf{r}}_i$:

$$\hat{\mathbf{e}}_i = f^e \left(\sum_{j \in \mathcal{N}^{etag}} \alpha_{i,j}^{etag} \mathbf{h}_j^{etag} \right) \quad (11)$$

$$\hat{\mathbf{r}}_i = f^r \left(\sum_{p \in \mathcal{N}^{rtag}} \alpha_{i,p}^{rtag} \mathbf{h}_p^{rtag} \right) \quad (12)$$

where f^e and f^r are the MLP aggregation functions for the influence of semantics on entities and relations, respectively, $\mathbf{h}_j^{etag}$ and $\mathbf{h}_p^{rtag}$ represent the entity semantic tag j and relation semantic tag p representations. $\alpha_{i,j}^{etag}$ and $\alpha_{i,p}^{rtag}$ represent the influence of the entity and relation semantic tag j and tag p on entity and relation i , which can be expressed as:

$$\alpha_{i,j}^{etag} = \frac{\exp(\phi(\mathbf{W}_3 \mathbf{h}_j^{etag} \parallel \mathbf{W}_4 \mathbf{e}_i))}{\sum_{k=1}^{|\mathcal{N}^{etag}|} \exp(\phi(\mathbf{W}_3 \mathbf{h}_k^{etag} \parallel \mathbf{W}_4 \mathbf{e}_i))} \quad (13)$$

$$\alpha_{i,p}^{rtag} = \frac{\exp(\phi(\mathbf{W}_5 \mathbf{h}_p^{rtag} \parallel \mathbf{W}_6 \mathbf{r}_i))}{\sum_{q=1}^{|\mathcal{N}^{rtag}|} \exp(\phi(\mathbf{W}_5 \mathbf{h}_q^{rtag} \parallel \mathbf{W}_6 \mathbf{r}_i))} \quad (14)$$

where $\mathbf{W}_3$, $\mathbf{W}_4$, $\mathbf{W}_5$, and $\mathbf{W}_6$ are learnable parameters, ϕ is the LeakyReLU activation function, and $|\mathcal{N}^{etag}|$ and $|\mathcal{N}^{rtag}|$ represent the total number of entity and relation semantic tags.

After integrating the semantic tags, the representations of entities and relations are updated via residual connections, following the approach proposed in [29]:

$$\mathbf{e}_i := \mathbf{e}_i + \beta_e \cdot \text{LeakyReLU}(\mathbf{W}_7 \hat{\mathbf{e}}_i) \quad (15)$$

$$\mathbf{r}_i := \mathbf{r}_i + \beta_r \cdot \text{LeakyReLU}(\mathbf{W}_8 \hat{\mathbf{r}}_i) \quad (16)$$

where β_e and β_r are values in the range $[0.1, 0.9]$, representing the influence of semantic tags on the entity and relation representations during the update. $\mathbf{W}_7$ and $\mathbf{W}_8$ are learnable transformation matrices.

4.3 Prediction and Optimization

Following [18], we adopt ConvTransE [35] as the decoder in our proposed model to predict the probability of future facts in a Temporal Knowledge Graph (TKG). Specifically, the probability score for each candidate entity e_o at time step t is defined as:

$$p(e_o | e_s, r, t) = \sigma(\mathbf{E}_t \text{ConvTransE}(\mathbf{e}_{s,t}, \mathbf{r}_t)) \quad (17)$$

where $\sigma(\cdot)$ is the sigmoid activation function, $\mathbf{e}_{o,t}$ is the representation of entity e_o at time t from the integrated entity representation matrix $\mathbf{E}_t \in \mathbb{R}^{|\mathcal{E}| \times d}$ (with each row corresponding to an entity), and $\text{ConvTransE}(\mathbf{e}_{s,t}, \mathbf{r}_t)$ generates the query embedding for the subject entity e_s and relation r at time t . To train the model, we maximize the prediction probability by minimizing the negative log-likelihood loss:

$$\mathcal{L} = - \sum_{t=0}^{T-1} \sum_{(e_s, r, e_o, t) \in \mathcal{G}_t} \log p(e_o | e_s, r, t) \quad (18)$$

where $\mathcal{G}_t = (e_s, r, e_o, t) \mid (e_s, r, e_o, t) \in \mathcal{G}$ denotes the set of ground-truth quadruples at time step t in the TKG, T is the number of time steps.

5 Experiments

In this section, we empirically evaluate the proposed DSTAG model on three standard TKG benchmark datasets: ICEWS14, ICEWS18, and ICEWS05-15. To provide a comprehensive evaluation of DSTAG's capabilities in TKGC, we focus on the following research questions:

- **Q1:** How does DSTAG perform in entities prediction compared to state-of-the-art baseline methods on benchmark datasets?
- **Q2:** How do key hyperparameters in DSTAG influence its predictive performance?
- **Q3:** What is the contribution of each component to DSTAG's overall performance?

Before addressing these questions, we detail the experimental settings, including datasets, evaluation metrics, and evaluation protocols.

5.1 Experimental Setup

5.1.1 Datasets. We evaluate DSTAG on three widely adopted TKG datasets: ICEWS14, ICEWS18, and ICEWS05-15. For a detailed description of the datasets, please refer to appendix A.2.

5.1.2 Evaluation Metrics. To evaluate the performance of our model on the TKGC task, we adopt two standard metrics: Mean Reciprocal Rank (MRR) and Hits at k (Hits@ k , where $k = 1, 3, 10$). For a more detailed description of these metrics, please refer to the appendix A.3.

5.1.3 Implementation Details. The proposed DSTAG model is implemented in Python using the PyTorch framework. To evaluate entity prediction performance, we use historical facts from time steps $t = 0$ to $T - 1$ in the TKG to predict missing entities at time T .

During training, the GCN component is configured with two layers and a hidden dimension of 200. The learning rate is set to 0.001, and model parameters are optimized using the Adam optimizer. We apply early stopping, terminating training if the validation MRR shows no improvement over 20 consecutive epochs. The model with the highest validation performance is subsequently tested on the test set.

For hyperparameter tuning, we employ a grid search strategy to optimize the parameters β_e and β_r in the SMP module, which

control the influence of semantic tags on entity and relation representations, respectively. Specifically, we first fix β_e at 0.1 and tune β_r over the range $[0.1, 0.9]$ with a step size of 0.2. After identifying the optimal β_r , we fix its value and tune β_e using the same range and step size.

5.1.4 Baselines. To evaluate the superiority of the proposed DSTAG model, we compare its entity prediction performance against 14 state-of-the-art baseline models, encompassing both static KGC and TKGC approaches. For baseline, we eliminated all temporal data from the benchmark datasets to suit static KGC models and documented their non-temporal performance. For TKGC baselines, we utilized publicly accessible implementations of REGCN [18], TiRGN [25], RETIA [33], LogCL [19] and TiRGN-ANEL [12] with their default parameter settings from GitHub. The results for other baselines were directly obtained from their original publications. For a more detailed description of the baseline, please refer to the appendix A.4.

5.2 Main Results (RQ1)

In Table 1, the best-performing results are highlighted in **bold**, while the second-best results are underlined. These results demonstrate the effectiveness of DSTAG in capturing semantic information for TKG completion.

The results show that the proposed DSTAG model consistently outperforms all baseline models across the three benchmark datasets, achieving the highest scores in MRR and Hits@ k ($k = 1, 3, 10$). Specifically, DSTAG improves the MRR by 8.64%, 9.81%, and 4.56% on ICEWS14, ICEWS18, and ICEWS05-15, respectively, compared to the second-best performing models. This superior performance can be attributed to DSTAG's innovative integration of semantic tag representations for entities and relations, along with its ability to capture higher-order interactions between them.

Static KGC models, such as ComplEx [36] and its temporal extension TNTComplEx [16], leverage complex-valued embeddings to capture relational patterns. However, they are inherently limited in modeling the dynamic temporal evolution of KGs. As a result, despite their effectiveness in handling static relations, their performance degrades significantly in scenarios involving temporal dynamics.

In contrast, TKGC models such as RE-NET [37] and REGCN [18] integrate RNNs and GCNs to capture temporal and structural dependencies. While these models can handle sequential data to some extent, they often fail to effectively model deep semantic representations of entities and the complex interactions among them, thereby limiting their expressive power for facts in continuously evolving TKGs.

More recent models like TiRGN [25] and TiRGN-ANEL [12] improve upon prior work by incorporating global contextual information and adaptive feature enhancement mechanisms. LLM-DA further introduces large language models to generate temporal rules for identifying query-relevant facts. Nonetheless, both approaches still fall short in fully capturing entity-specific semantics and intricate relational dependencies, which are critical for high-fidelity temporal reasoning.

Other models, such as LogCL [19] and CENET [38], attempt to enhance performance by incorporating attention mechanisms and

Table 1: ICEWS14, ICEWS18, and ICEWS05-15. Best results are in bold.

Model	ICEWS14				ICEWS05-15				ICEWS18			
	MRR	Hit@1	Hit@3	Hit@10	MRR	Hit@1	Hit@3	Hit@10	MRR	Hit@1	Hit@3	Hit@10
ComplEx (PMLR'2016)	31.79	22.91	36.13	51.43	22.55	14.22	25.92	40.96	18.76	11.39	21.59	25.73
ConvE (AAAI'2018)	31.23	21.20	36.13	50.37	30.40	20.21	36.05	49.96	24.16	15.47	23.52	44.32
R-GCN (ESWC'2018)	28.03	19.42	31.95	47.46	27.13	18.83	30.41	43.16	15.05	8.13	16.49	29.00
TNTComplEx (ICLR'2020)	30.02	23.33	33.94	49.11	27.52	19.50	30.80	42.85	21.21	13.26	24.02	36.89
RE-NET (ACL'2020)	38.21	28.43	39.51	54.11	42.77	30.89	47.77	63.29	28.83	19.07	32.44	47.53
xERTE (ICLR'2021)	40.69	32.65	44.96	57.26	46.57	37.76	52.31	68.87	29.28	21.01	33.46	45.56
REGCN (SIGIR'2021)	41.75	31.59	46.43	61.46	46.44	35.99	52.08	66.45	32.2	22.08	36.25	51.98
TiRGN (IJCAI'2022)	43.67	33.09	48.11	62.21	49.36	38.73	55.27	69.72	33.26	22.87	37.58	53.64
CENET (AAAI'2023)	41.25	32.23	43.23	58.16	47.05	37.16	49.93	67.57	29.63	19.87	33.63	48.12
RETIA (ICDE'2023)	42.77	32.29	47.77	63.66	47.29	36.69	52.90	67.77	32.34	22.23	36.48	52.97
HTCCN (ACL'2024)	45.39	36.58	50.84	67.29	51.94	40.32	57.79	73.53	35.63	24.90	39.26	56.22
LogCL (ICDE'2024)	<u>47.56</u>	<u>37.32</u>	<u>54.65</u>	<u>71.08</u>	<u>57.20</u>	<u>46.18</u>	<u>63.89</u>	<u>78.19</u>	34.86	23.36	39.82	<u>57.56</u>
LLM-DA (NeurIPS'2024)	47.10	36.93	52.59	67.08	52.12	41.59	58.61	72.78	<u>35.68</u>	<u>25.52</u>	<u>40.31</u>	57.02
TiRGN-ANEL (WWW'2025)	44.74	34.4	49.82	64.66	50.59	39.69	56.67	71.27	34.04	23.53	38.41	54.71
DSTAG (ours)	51.67	40.76	57.37	72.73	59.81	48.85	66.69	80.39	39.18	27.52	43.61	61.85
Δ improve	8.64%	9.22%	4.98%	2.32%	4.56%	5.78%	4.38%	2.81%	9.81%	7.84%	8.19%	7.45%

contrastive learning strategies. However, they still lack depth in semantic modeling and are often sensitive to noise, which undermines their robustness. RETIA [33] and HTCCN [39] contribute to temporal reasoning by modeling historical facts and capturing time series dependencies. Nevertheless, they exhibit limitations in representing higher-order interaction patterns between entities and relations.

In contrast, DSTAG demonstrates superior performance in entity prediction tasks for TKGc by effectively encoding rich semantic features and capturing complex higher-order interactions, thereby addressing the limitations observed in prior models.

5.3 Sensitivity Analysis (RQ2)

To investigate the impact of residual-connection weights on entity–relation fusion and its subsequent effect on entity prediction performance, we conduct a comprehensive sensitivity analysis of the parameters β_e and β_r across three benchmark datasets.

The optimal parameter combination is obtained through a two-stage tuning process. In the first stage, β_e is fixed at 0.1 while β_r is varied over the range [0.1, 0.9] with a step size of 0.2. The configuration yielding the best performance is selected as the optimal β_r . In the second stage, this β_r is held constant while β_e is tuned using the same search range and step size.

As shown in Figure 2, on the ICEWS14 dataset, the model achieves optimal performance when $\beta_e = 0.1$ and $\beta_r = 0.3$, reaching an MRR of 51.67 and a Hit@1 of 40.76. This suggests that a lower β_e effectively constrains the update magnitude of entity embeddings, helping preserve original feature representations, while a moderate β_r enhances the expressive power of relation embeddings. In comparison, a suboptimal combination (e.g., $\beta_e = 0.3$, $\beta_r = 0.5$) leads to a decreased MRR of 51.02, indicating a noticeable drop in prediction accuracy.

As illustrated in Figure 3, on the ICEWS18 dataset, the optimal parameter setting is $\beta_e = 0.1$ and $\beta_r = 0.1$, yielding an MRR of 39.18 and a Hit@1 of 27.52. Given ICEWS18's shorter temporal span and simpler event structures, a lower β_r prevents over-adjustment

of relation embeddings. Sensitivity analysis further reveals that increasing β_r to 0.5 causes the MRR to drop by approximately 1.77%, suggesting that excessive tuning of relation embeddings can lead to overfitting.

As depicted in Figure 4, the best performance on the ICEWS05-15 dataset is achieved with $\beta_e = 0.1$ and $\beta_r = 0.5$, where the MRR reaches 59.81 and Hit@1 is 48.85. Owing to the dataset's longer time span and greater event complexity, higher values of β_r contribute to capturing long-term dependencies and modeling complex relational patterns more effectively.

5.4 Ablation Study (RQ3)

To evaluate the contribution of each component in the DSTAG model to entity prediction performance, we conducted ablation experiments on three benchmark datasets: ICEWS14, ICEWS18, and ICEWS05-15. MRR and Hit@k(1,3,10) is used as the evaluation metric. In these experiments, we sequentially remove key components of the model—including entity and relation semantic tags (ST), the relation graph (RG), entity semantic tags (EST), and relation tags (RT)—to assess their individual impact on overall performance. The corresponding experimental results are presented in Table 2, highlighting the contribution of each architectural component to the DSTAG model's overall effectiveness in temporal reasoning.

- **Impact of Semantic Tags:** To evaluate the role of semantic tags, we removed both entity and relation semantic tags ("w/o ST") and replaced them with randomly initialized embeddings. These embeddings were then propagated through the entity and relation graphs for entity prediction. The results demonstrate a substantial drop in performance, confirming that semantic tags provide essential contextual knowledge that enhances the model's capacity to capture semantic nuances and improve prediction accuracy.
- **Impact of the Relation Graph:** In the "w/o RG" setting, the relation graph was excluded, and relation embeddings were learned as independent parameters, combined with the entity semantic tag graph. The performance degradation observed in this setting

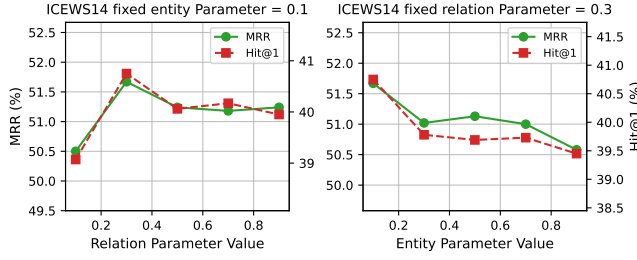


Figure 2: Performance of DSTAG model under different parameter value on ICEWS14 dataset.

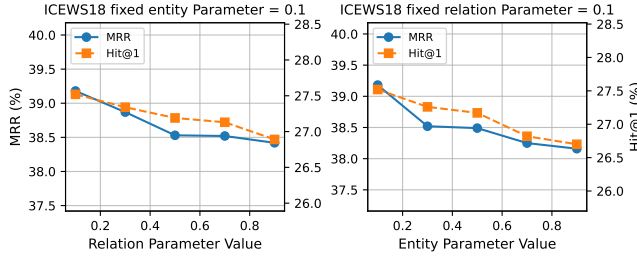


Figure 3: Performance of DSTAG model under different parameter value on ICEWS18 dataset.

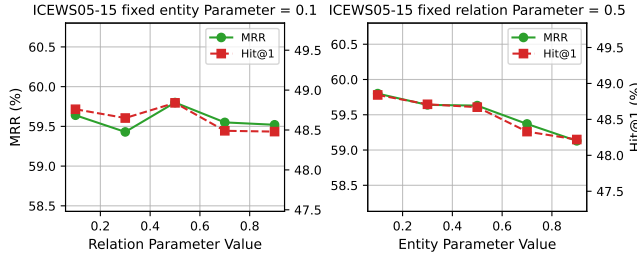


Figure 4: Performance of DSTAG model under different parameter value on ICEWS05-15 dataset.

highlights the critical role of the relation graph in modeling complex relational semantics and high-order interactions, especially in temporally evolving contexts.

- **Impact of Entity Semantic Tags:** In the "w/o EST" experiment, the entity semantic tags were removed, and the model relied solely on the entity graph and the relation semantic tag graph for entity prediction. The results show a consistent decline in performance across all datasets compared to the full DSTAG model, demonstrating the effectiveness of incorporating entity semantic tags. These tags enrich the semantic representation of entities, thereby contributing to more accurate prediction.
- **Impact of Relation Semantic Tags:** In the "w/o RST" experiment, the relation semantic tags were removed to evaluate their effect on model performance. The results demonstrate that incorporating relation semantic tags significantly enhances predictive accuracy, particularly on datasets with high relational complexity.

Table 2: Performance of different modules of DSTAG on various datasets.

Dataset	Ablation	MRR	H@1	H@3	H@10
ICEWS14	w/o ST	49.83	38.87	55.75	70.33
	w/o RG	50.70	39.73	56.84	71.73
	w/o EST	50.49	39.22	56.68	71.80
	w/o RST	50.69	39.81	56.30	71.23
	DSTAG	51.67	40.76	57.37	72.73
ICEWS05-15	w/o ST	58.87	47.93	65.77	79.31
	w/o RG	59.14	48.31	65.93	79.48
	w/o EST	59.17	48.53	66.23	80.06
	w/o RST	59.50	48.56	66.45	80.18
	DSTAG	59.81	48.45	66.69	80.39
ICEWS18	w/o ST	38.46	26.77	43.73	60.61
	w/o RG	38.16	26.46	43.13	60.41
	w/o EST	38.74	27.17	43.96	61.27
	w/o RST	38.51	26.87	43.78	61.03
	DSTAG	39.18	27.52	43.61	61.85

These tags play a crucial role in enriching relational representations, thereby enabling the model to better capture intricate relationship patterns.

6 Conclusion and Future Work

Accurate TKGC requires models to fully capture historical information relevant to a given query. For this purpose, we propose DSTAG, a novel entity prediction model that combines a dual-graph convolutional network with semantic tag enhancement. DSTAG constructs entity and relation graphs based on temporal facts, leverages a LLM to generate contextual semantic tags for entities and relations, and incorporates a Semantic Tag Network to update and propagate these tags, thereby enriching the semantic expressiveness of the model. Comprehensive evaluations across three authentic TKG datasets reveal that DSTAG markedly surpasses existing baselines. However, the current approach models semantic tags only at the current timestamp, without capturing the dynamic evolution of semantic tags over time or contextual semantics tied to specific historical moments. Moreover, while LLMs are used to generate high-quality tags, their potential for dynamic temporal reasoning remains underexplored.

In future work, we plan to design a dynamic semantic tag generation mechanism that fully exploits the reasoning capabilities of LLMs. This mechanism will aim to model the higher-order evolution of semantic information across time and differentiate its influence on the current entity and relation representations. This opens promising directions for advancing semantic-aware and temporally adaptive reasoning in TKGs.

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References

- [1] Yuchao Zhang, Xiangjie Kong, Zhehui Shen, Jianxin Li, Qihua Yi, Guojiang Shen, and Bo Dong. A survey on temporal knowledge graph embedding: Models and applications. *Knowledge-Based Systems*, page 112454, 2024.
- [2] Shangfei Zheng, Xiaotong Nie, Wei Yuan, Liang Qu, Xiangjie Kong, Yuchao Zhang, and Jian Hou. Context-aware adaptive reinforcement learning for multi-hop knowledge graph reasoning in few-shot scenarios. *Knowledge-Based Systems*, page 114799, 2025.
- [3] Shangfei Zheng, Hongzhi Yin, Tong Chen, Quoc Viet Hung Nguyen, Wei Chen, and Lei Zhao. Do as i can, not as i get: Topology-aware multi-hop reasoning on multi-modal knowledge graphs. *IEEE Transactions on Knowledge and Data Engineering*, 2025.
- [4] Xing Tang, Ling Chen, Hongyu Shi, and Dandan Lyu. Dhyper: a recurrent dual hypergraph neural network for event prediction in temporal knowledge graphs. *ACM Transactions on Information Systems*, 42(5):1–23, 2024.
- [5] Wei Chen, Huaiyu Wan, Shengnan Guo, Haoyu Huang, Shaojie Zheng, Jiamu Li, Shuohao Lin, and Youfang Lin. Building and exploiting spatial-temporal knowledge graph for next poi recommendation. *Knowledge-Based Systems*, 258:109951, 2022.
- [6] Hengchang Hu, Wei Guo, Xu Liu, Yong Liu, Ruiming Tang, Rui Zhang, and Min-Yen Kan. User behavior enriched temporal knowledge graphs for sequential recommendation. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*, pages 266–275, 2024.
- [7] Xiaoman Zhang, Linlin Zhang, Xuehua Bi, Guanglei Yu, Ruyi Cao, and Yang Chen. Tkmbr: Temporal knowledge graph-based multi-behavior recommendation. In *2024 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2024.
- [8] Xinying Qian, Ying Zhang, Yu Zhao, Baohang Zhou, Xuhui Sui, Li Zhang, and Kehui Song. Timer4: Time-aware retrieval-augmented large language models for temporal knowledge graph question answering. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 6942–6952, 2024.
- [9] Yifu Gao, Linbo Qiao, Zhigang Kan, Zhihua Wen, Yongquan He, and Dongsheng Li. Two-stage generative question answering on temporal knowledge graph using large language models. *arXiv preprint arXiv:2402.16568*, 2024.
- [10] Lihui Liu, Zihao Wang, Dawei Zhou, Ruijie Wang, Yuchen Yan, Bo Xiong, Sihong He, Kai Shu, and Hanghang Tong. Transnet: Transfer knowledge for few-shot knowledge graph completion. *arXiv preprint arXiv:2504.03720*, 2025.
- [11] Mingyang Chen, Wen Zhang, Wei Zhang, Qiang Chen, and Huajun Chen. Meta relational learning for few-shot link prediction in knowledge graphs. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 4217–4226, 2019.
- [12] Yuchao Zhang, Xiangjie Kong, Kailun Ye, Guojiang Shen, and Shangfei Zheng. Tackling sparse facts for temporal knowledge graph completion. In *THE WEB CONFERENCE 2025*, WWW '25, page 3561–3570.
- [13] Tingsong Jiang, Tianyu Liu, Tao Ge, Lei Sha, Sujian Li, Baobao Chang, and Zhifang Sui. Encoding temporal information for time-aware link prediction. In *Proceedings of the 2016 conference on empirical methods in natural language processing*, pages 2350–2354, 2016.
- [14] Ali Sadeghian, Mohammadreza Armandpour, Anthony Colas, and Daisy Zhe Wang. Chronor: Rotation based temporal knowledge graph embedding. In *Proceedings of the AAAI conference on artificial intelligence*, volume 35, pages 6471–6479, 2021.
- [15] Lifan Lin and Kun She. Tensor decomposition-based temporal knowledge graph embedding. In *2020 IEEE 32nd International conference on tools with artificial intelligence (ICTAI)*, pages 969–975. IEEE, 2020.
- [16] Timothée Lacroix, Guillaume Obozinski, and Nicolas Usunier. Tensor decompositions for temporal knowledge base completion. *arXiv preprint arXiv:2004.04926*, 2020.
- [17] Chengjin Xu, Yung-Yu Chen, Mojtaba Nayyeri, and Jens Lehmann. Temporal knowledge graph completion using a linear temporal regularizer and multivector embeddings. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 2569–2578, 2021.
- [18] Zixuan Li, Xiaolong Jin, Wei Li, Saiping Guan, Jiafeng Guo, Huawei Shen, Yuanzhuo Wang, and Xueqi Cheng. Temporal knowledge graph reasoning based on evolutionary representation learning. In *Proceedings of the 44th international ACM SIGIR conference on research and development in information retrieval*, pages 408–417, 2021.
- [19] Wei Chen, Huaiyu Wan, Yuting Wu, Shuyuan Zhao, Jiayao Cheng, Yuxin Li, and Youfang Lin. Local-global history-aware contrastive learning for temporal knowledge graph reasoning. In *2024 IEEE 40th International Conference on Data Engineering (ICDE)*, pages 733–746. IEEE, 2024.
- [20] Mengqi Zhang, Yuwei Xia, Qiang Liu, Shu Wu, and Liang Wang. Learning latent relations for temporal knowledge graph reasoning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12617–12631, 2023.
- [21] Xing Tang and Ling Chen. Gtrl: An entity group-aware temporal knowledge graph representation learning method. *IEEE Transactions on Knowledge and Data Engineering*, 36(9):4707–4721, 2023.
- [22] Zhen Han, Peng Chen, Yunpu Ma, and Volker Tresp. Explainable subgraph reasoning for forecasting on temporal knowledge graphs. In *International conference on learning representations*, 2020.
- [23] Zhiwen Xie, Runjie Zhu, Jin Liu, Guangyou Zhou, and Jimmy Xiangji Huang. Targat: A time-aware relational graph attention model for temporal knowledge graph embedding. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 31:2246–2258, 2023.
- [24] Woojeong Jin, Meng Qu, Xisen Jin, and Xiang Ren. Recurrent event network: Autoregressive structure inference over temporal knowledge graphs. *arXiv preprint arXiv:1904.05530*, 2019.
- [25] Yujia Li, Shiliang Sun, and Jing Zhao. Tirgn: Time-guided recurrent graph network with local-global historical patterns for temporal knowledge graph reasoning. In *IJCAI*, pages 2152–2158, 2022.
- [26] Yushan Liu, Yunpu Ma, Marcel Hildebrandt, Mitchell Joblin, and Volker Tresp. Tloglc: Temporal logical rules for explainable link forecasting on temporal knowledge graphs. In *Proceedings of the AAAI conference on artificial intelligence*, volume 36, pages 4120–4127, 2022.
- [27] Yuwei Xia, Ding Wang, Qiang Liu, Liang Wang, Shu Wu, and Xiaoyu Zhang. Chain-of-history reasoning for temporal knowledge graph forecasting. *arXiv preprint arXiv:2402.14382*, 2024.
- [28] Qihong Pan, Limin Yao, Guojiang Shen, Xiao Han, Yichuan Chen, and Xiangjie Kong. Leveraging temporal validity of rules via llms for enhanced temporal knowledge graph reasoning. *Knowledge-Based Systems*, page 114094, 2025.
- [29] Zifeng Ding, Jingpei Wu, Bailan He, Yunpu Ma, Zhen Han, and Volker Tresp. Few-shot inductive learning on temporal knowledge graphs using concept-aware information. *arXiv preprint arXiv:2211.08169*, 2022.
- [30] Mehrnoosh Mirtaheri, Ryan A Rossi, Sungchul Kim, Kanak Mahadik, Tong Yu, Xiang Chen, and Mohammad Rostami. Tackling long-tail entities for temporal knowledge graph completion. In *Companion Proceedings of the ACM on Web Conference 2024*, pages 497–500, 2024.
- [31] Ling Chen, Xing Tang, Weiqi Chen, Yuntao Qian, Yansheng Li, and Yongjun Zhang. Dacha: A dual graph convolution based temporal knowledge graph representation learning method using historical relation. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 16(3):1–18, 2021.
- [32] Shuxian Huang, Ye Wang, Kai Chen, and Yan Jia. Temporal relational context learning for extrapolation reasoning on temporal knowledge graphs. In *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 6430–6434. IEEE, 2024.
- [33] Kangzheng Liu, Feng Zhao, Guandong Xu, Xianzhi Wang, and Hai Jin. Retia: relation-entity twin-interact aggregation for temporal knowledge graph extrapolation. In *2023 IEEE 39th international conference on data engineering (ICDE)*, pages 1761–1774. IEEE, 2023.
- [34] Zixuan Li, Zhongni Hou, Saiping Guan, Xiaolong Jin, Weihua Peng, Long Bai, Yajuan Lyu, Wei Li, Jiafeng Guo, and Xueqi Cheng. Hismatch: Historical structure matching based temporal knowledge graph reasoning. *arXiv preprint arXiv:2210.09708*, 2022.
- [35] Chao Shang, Yun Tang, Jing Huang, Jinbo Bi, Xiaodong He, and Bowen Zhou. End-to-end structure-aware convolutional networks for knowledge base completion. In *Proceedings of the AAAI conference on artificial intelligence*, volume 33, pages 3060–3067, 2019.
- [36] Théo Trouillon, Johannes Welbl, Sebastian Riedel, Éric Gaussier, and Guillaume Bouchard. Complex embeddings for simple link prediction. In *International conference on machine learning*, pages 2071–2080. PMLR, 2016.
- [37] Woojeong Jin, Meng Qu, Xisen Jin, and Xiang Ren. Recurrent event network: Autoregressive structure inference over temporal knowledge graphs. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6669–6683, 2020.
- [38] Yi Xu, Junjie Ou, Hui Xu, and Luoyi Fu. Temporal knowledge graph reasoning with historical contrastive learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 4765–4773, 2023.
- [39] Tingxuan Chen, Jun Long, Liu Yang, Zidong Wang, Yongheng Wang, and Xiongnan Jin. Htcn: temporal causal convolutional networks with hawks process for extrapolation reasoning in temporal knowledge graphs. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 4056–4066, 2024.
- [40] Tim Dettmers, Pasquale Minervini, Pontus Stenetorp, and Sebastian Riedel. Convolutional 2d knowledge graph embeddings. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.
- [41] Michael Schlichtkrull, Thomas N Kipf, Peter Bloem, Rianne Van Den Berg, Ivan Titov, and Max Welling. Modeling relational data with graph convolutional networks. In *The semantic web: 15th international conference, ESWC 2018, Heraklion, Crete, Greece, June 3–7, 2018, proceedings 15*, pages 593–607. Springer, 2018.

[42] Jiapu Wang, Sun Kai, Linhao Luo, Wei Wei, Yongli Hu, Alan Wee-Chung Liew, Shirui Pan, and Baocai Yin. Large language models-guided dynamic adaptation for temporal knowledge graph reasoning. *Advances in Neural Information Processing Systems*, 37:8384–8410, 2024.

A Appendix

A.1 Symbol Explanation

The important symbols used in this article and their corresponding meanings are shown in Table 3.

Table 3: List of Main Notations.

Notation	Description
$\mathcal{G}$	Temporal knowledge graph
$\mathcal{E}$	The set of entities
$\mathcal{R}$	The set of relations
$\mathcal{F}$	The set of facts
$\mathcal{T}$	The set of timestamps
$\mathcal{G}_t$	The subgraph at time t
$\mathcal{G}_e^t$	The entity graph at time t
$\mathcal{G}_r^t$	The relationship graph at time t
$\mathcal{V}_r^t$	The set of nodes in the relationship graph at time t
$\mathcal{N}^{etag}$	The set of entity semantic tags
$\mathcal{N}^{rtag}$	The set of relation semantic tags
$\mathcal{N}_j^{etag}$	The j -th entity semantic tag
$\mathcal{N}_p^{rtag}$	The p -th relation semantic tag
Δt	The time interval
$\mathbf{e}_t$	The entity embedding at time t
$\mathbf{E}_t$	The entity embedding matrices at time t
$\mathbf{R}_t$	The relation embedding matrices at time t
$\mathcal{N}_{r_i,t}$	The set of neighbor nodes of r_i
$\mathbf{W}$	The learnable parameter matrix
$\mathbf{g}_{R_t}$	A learnable gating vector
α_j^{ei}	The relevance of the entity representations to the entity tag j
α_p^{ri}	The relevance of the relation representations to the entity tag p
$\mathbf{h}_j^{etag}$	The final j -th entity semantic tag representation
$\mathbf{h}_p^{rtag}$	The final p -th relation semantic tag representation

A.2 Dataset Statistics.

All datasets are derived from the Integrated Crisis Early Warning System (ICEWS), containing timestamped geopolitical event records. ICEWS14 and ICEWS18 correspond to events from 2014 and 2018, respectively, while ICEWS05-15 spans from 2005 to 2015. To avoid temporal information leakage, each dataset is chronologically split into training (80%), validation (10%), and test (10%) sets. Table 1 summarizes key dataset statistics.

A.3 Evaluation Metrics.

MRR measures the average reciprocal rank of the correct entity in the predicted ranking, reflecting the model’s ability to assign high ranks to ground-truth entities. Hits@ k indicates the proportion of queries for which the correct entity appears in the top- k predictions, thereby assessing the model’s accuracy within a ranked shortlist.

Table 4: Common TKGE benchmarks and their attributes

Datasets	ICEWS14	ICEWS05-15	ICEWS18
Entities	7128	10488	23033
Relations	230	251	256
Facts	90730	479329	468558
Train Facts	74845	368868	373018
Time Span	365	4017	304

Both metrics are computed under a time-aware filtered setting, where only candidate entities that are valid at the same timestamp as the query are considered. This setting, consistent with the TiRGN model [25], ensures a fair and meaningful comparison. Higher MRR and Hits@ k values indicate better performance. All results are reported as percentages.

A.4 Baselines Details

The static KGC baselines include ComplEx [36], ConvE [40], and R-GCN [41], selected for their robust performance in modeling entity-relation interactions. The TKGC baselines comprise TNTComplEx [16], RE-NET [37], xERTE [22], REGCN [18], TiRGN [25], GTRL [21], CENET [38], RETIA [33], LogCL [19], HTCCN [39], LLM-DA [42] and TiRGN-ANEL [12] chosen for their advanced capabilities in capturing temporal dynamics and historical context in TKGs.

KGC Models

- ComplEx [36] uses complex space embeddings, specifically designed to capture symmetric and antisymmetric relationships between entities. Through the properties of complex vectors, ComplEx better represents the bidirectional nature of relations.
- ConvE [40] uses Convolutional Neural Networks (CNNs) to capture the complex interactions between entities and relations. Through convolution operations, ConvE efficiently models the nonlinear relationships between high-dimensional features.
- R-GCN [41] model multi-relational interactions between entities.

TKGC Models

- TNTComplEx [16] incorporates the time dimension into the complex space embeddings to efficiently capture the evolving complex relationships of entities over time.
- RE-NET [37] combines RNN and R-GCN. The RNN captures the temporal dependencies of entities, while the R-GCN captures the interaction dependencies between entities and relations, enabling dual modeling of temporal and structural information.
- xERTE [22] is an interpretable model that uses a subgraph attention mechanism to predict new events by utilizing historically relevant facts.
- REGCN [18] employs an RNN to capture temporal relationships among entities and integrates Graph Convolutional Networks (GCN) to represent spatial interactions between entities. It also captures the dynamic evolution of entity and relation representations over time.
- TiRGN [25] is an extension of REGCN, which incorporates historical entity-relation information as additional inputs for entity and relation representations, enhancing the model’s ability to perceive historical context.

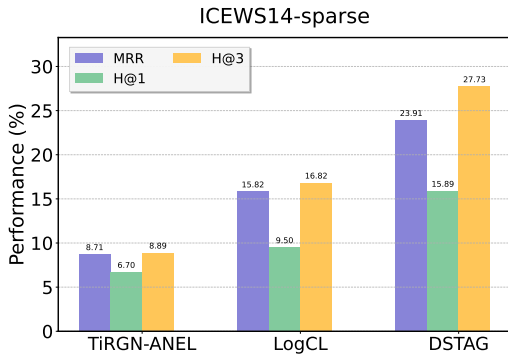


Figure 5: Extreme sparsity analysis.

- CENET [38] is a contrastive learning-based model that improves prediction accuracy by distinguishing the importance of dependencies and non-dependencies of historical facts in the current prediction task.
- RETIA [33] uses dual hyper-relational subgraphs to model historical facts and employs GCN to capture the interaction dependencies between entities within subgraphs, enhancing the representation of complex historical relationships.
- HTCCN [39] uses a temporal causal convolutional network to model sequential dependencies among facts and leverages a Hawkes process to capture the occurrence dynamics of facts.
- LLM-DA [42] uses Large Language Models to capture temporal rules from the data, continuously updates these rules with new events, and uses the generated temporal rules to select the most relevant facts.

- LogCL [19] utilizes attention mechanisms to distinguish the relevance of temporal subgraphs and queries, dynamically adjusting entity representations based on different levels of relevance.
- TiRGN-ANEL [12] is an extension of TiRGN that adds an adaptive enhancement layer, dynamically supplementing additional features based on the interaction information of entities in the TKG, further improving the expressiveness of entities.

A.5 Extreme Sparsity Analysis (RQ1)

In real-world applications, temporal knowledge graphs frequently encounter extreme sparsity issues, where new entities emerge frequently or existing entities are associated with only a very limited number of historical facts. This places higher demands on the model's generalization ability and semantic reasoning capabilities. Particularly in cold-start or sudden-event scenarios, traditional methods that rely heavily on sufficient historical structural information often struggle to produce reliable predictions.

To systematically evaluate the robustness and effectiveness of the proposed DSTAG model under such extreme sparsity conditions, this chapter conducts further 3-shot entity prediction experiments on the ICEWS14 dataset. Specifically, in this setting, each test entity and validation entity retains no more than 3 reference triples within the historical time window, significantly reducing the availability of structural and temporal information. This setup ensures task learnability while more closely aligning with real-world application needs for low-frequency entities and emerging events. DSTAG is compared with several representative and competitive strong baseline models, including TiRGN-ANEL and LogCL. These methods address data sparsity from different perspectives—neighborhood enhancement, contrastive learning, and semantic alignment—making them highly representative and competitive. The experimental results are shown in the figure 5.